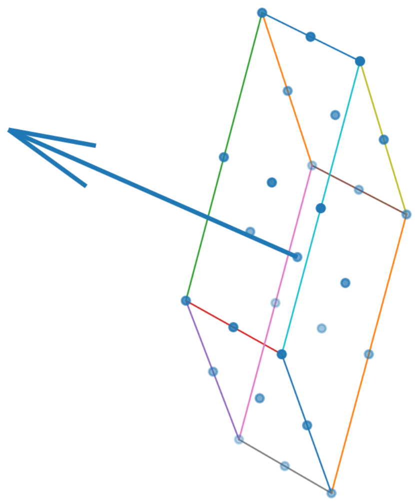
K-FIELD

Literature Review of Drop-Tower Tests with Rotating Flywheels, Gyroscopes, and Oscillating Masses

Cell3 geometry and observer-indexed field-response direction



The source-locked Cell3 geometry and the observer-indexed field-response direction used throughout this report.

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Frontispiece: authoritative base cell, planes, and all nodes

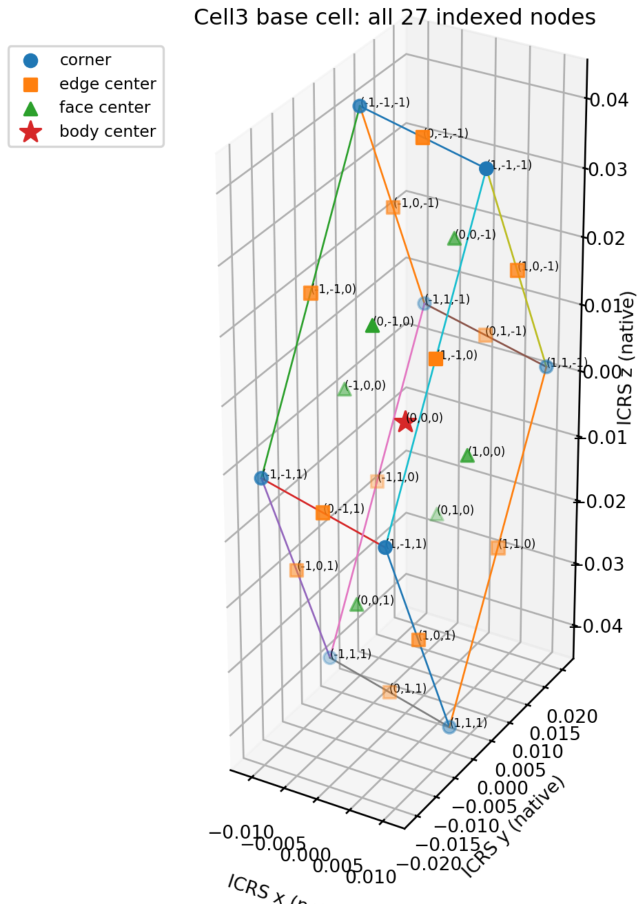

Figure 1. All 27 Cell3 nodes, indexed by the exact integer address (i,j,k) and plotted from the source JSON coordinates. Corner, edge-center, face-center, and body-center classes are distinguished.

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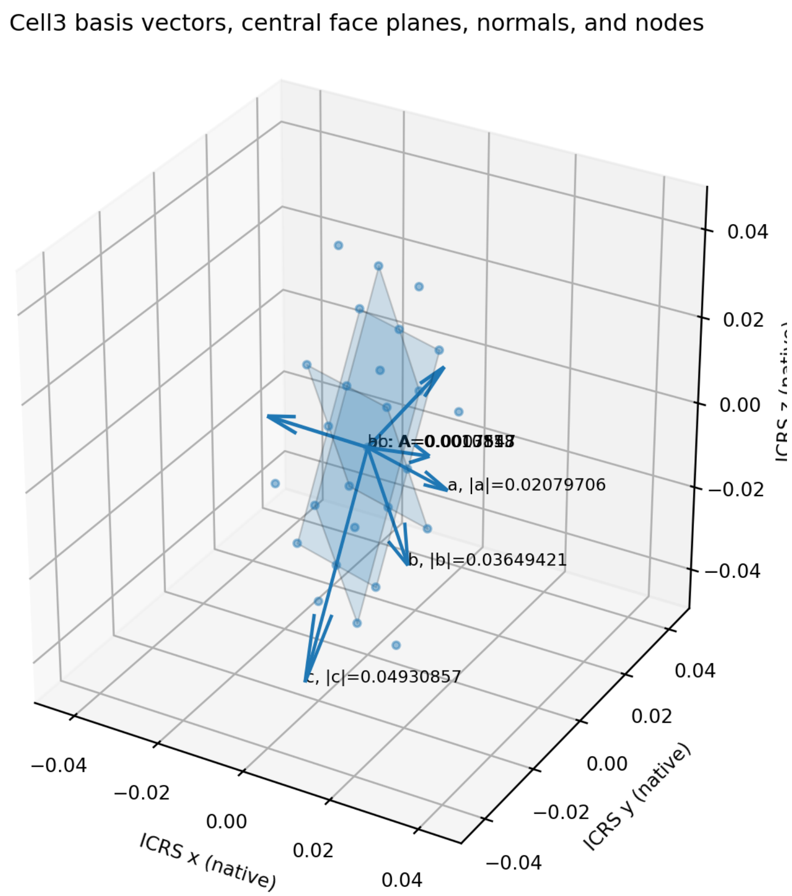

Figure 2. Dimensionally scaled basis vectors a, b, and c; central ab, ac, and bc planes; source face normals; and all Cell3 nodes.

The plotted node coordinates are not illustrative placements. Each node is reconstructed exactly as r(i,j,k) = 0.5 B [i j k]^T, where i,j,k are in {-1,0,+1} and B is the source basis matrix with columns a, b, and c.

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Contents

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1. Purpose, scope, and strict computation rule

This report examines publicly available drop-tower, free-fall, precision-weighing, gyroscope, flywheel, and inertial-sensor research for experiments structurally similar to the proposed K-Field drop test. It preserves the complete substantive findings of the preceding review while applying a stricter numerical rule: K-Field quantities are calculated only from explicitly reported mechanical inputs. Missing rotor mass, rotor geometry, RPM, fall distance, fall time, epoch, laboratory position, or axis orientation is not replaced by an assumed value.

The literature contains several close precedents, but no located study combines a freely falling horizontal-axis rotor, direct two-axis lateral position and velocity measurements, clockwise/counterclockwise/nonrotating controls, exact UTC and local-sidereal timestamps, and repeated observations designed to resolve an approximately 12-hour modulation. Most rotating-body studies measured apparent weight or vertical free-fall acceleration. The closest horizontal-axis reports did not directly measure lateral deflection, and the best controlled interferometric free-fall null was nearly insensitive to horizontal motion [1].

The report therefore separates three categories: (1) source-reported measurements; (2) K-Field maximum envelopes that can be computed from complete mechanical data; and (3) realized K-Field vectors, which additionally require the observer state, epoch, laboratory coordinates, and spin-axis direction.

Requested quantity	Minimum admissible inputs	Treatment when incomplete
Maximum acceleration	Explicit RPM, rotor geometry, field-response lock, and measured m_r/M or an explicit standalone-rotor condition	Not computed if RPM, geometry, or the inertial-mass relationship is absent
Maximum force	RPM, rotor geometry, measured active rotating mass m_r, and field-response lock	No density or mass is inferred
Drop displacement	Admissible acceleration plus measured fall distance or fall time	No displacement inferred from facility height alone
Realized vector	All above plus epoch, latitude, longitude, altitude, axis attitude, and spin sign	Only a maximum envelope is permitted when directional state is missing

Computational status is attached to every study so that absence of a number is informative rather than silently repaired.

2. K-Field kinematic foundation

The supplied rotating-mass reduction is

```
F_K = (kappa_F m_r omega / 2) (n x g), g = G_phase V_observer
```

Here m_r is the participating rotating mass, omega is signed angular speed, n is the spin-axis unit vector, and g is the observer-indexed field-response vector. If the rotor is attached to a larger freely falling assembly of total inertial mass M, the center-of-mass acceleration is

```
a_CM = F_K / M = (m_r/M) (kappa_F omega / 2) (n x g).
```

At perpendicular alignment, |n x g| = |g|. Defining q_g = kappa_F |g| / 2 gives F_max = m_r q_g omega and a_CM,max = (m_r/M) q_g omega. For a vacuum drop from rest through vertical height h, T = sqrt(2h/g0) and a constant transverse acceleration produces d = 0.5 a_CM T^2 = a_CM h/g0.

Mass and thickness cancel only for a standalone rotor that is itself the entire freely falling inertial body, so that m_r = M. If a thinner rotor is carried by a fixed-mass frame, release mechanism, optical reflector, or housing, m_r/M decreases and the predicted assembly deflection decreases in the same proportion. This distinction is essential when interpreting prior experiments and when designing a thin-rotor test.

Quantity	Source-reference lock	Controlling observer-state lock
kappa_F	3.07405e-11	same coupling coefficient
|g| (m/s)	9.30825e+05	8.25840e+05
q_g (N per kg per rad/s)	1.43070205e-05	1.26933745e-05
Velocity basis	JSON V_E vector	341.071130 km/s, RA 167.143423 deg, Dec -7.655565 deg
Use	Continuity with prior calculations and historical maximum envelopes	Realized drop-tower direction and baseline computations

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3. Observer-State Baseline and Directional Gravity Handling

The controlling observer baseline is SGA-2020 / ICRS / J2000 at 2026-06-17T15:06:11Z, latitude 42.129 deg, longitude -83.144 deg, altitude 0 m. The observer speed is 341.071130 km/s at RA 167.143423 deg and Dec -7.655565 deg. The CMB-closure direction is (-0.97077086, 0.20734801, -0.12087493).

All realized vectors in this report are formed in ICRS/J2000. The laboratory east-north-up basis is rotated into ICRS with Greenwich mean sidereal angle and IAU-1976 precession, matching the current MATLAB stack. The due-north spin axis is transformed at every sampled epoch, the cross product is evaluated in ICRS, and the result is transformed back into the local laboratory frame. UT1-UTC, nutation, and polar motion are not supplied; they are therefore not invented. The numerical realization uses UTC as the rotation-time argument and states that approximation explicitly.

G-derived rows use the observer-indexed chain g = G_phase V_observer. Source-reference rows are retained only as labeled envelopes to preserve continuity with earlier work; they are not pooled with observer-realized rows.

Observer field item	Value
Epoch	2026-06-17T15:06:11Z
Location	42.129 deg N, 83.144 deg W, 0 m
V_observer ICRS (m/s)	-3.295566967e+05, 7.521574118e+04, -4.543667771e+04
g_observer ICRS (m/s)	-7.928669020e+05, -1.093953161e+05, 2.034866600e+05
|g_observer| (m/s)	8.258402268e+05
q_eff	1.269337446e-05
f_obs	1.44774
beta_obs^2	1.55574e-06
zeta_R	0.320892, 0.374911, 0.838466

4. Cell3 base geometry and plane structure

The Cell3 basis is preserved as an exact geometric lock. The three basis-vector lengths, included angles, face areas, primitive volume, and plane normals are listed below. These values generate every coordinate in Figures 1 and 2 and the complete node appendix.

Plane	Edge lengths	Included angle	Area	Unit normal in ICRS
ab	a=0.0207970588; b=0.0364942051	83.933483012 deg	0.000754721790	0.194092201, 0.913177169, 0.358379235
ac	a=0.0207970588; c=0.0493085659	80.448130018 deg	0.001011255758	0.089052594, 0.866576769, -0.491033948
bc	b=0.0364942051; c=0.0493085659	50.363231814 deg	0.001385784432	-0.995965190, 0.083602163, -0.032619291

```
Primitive Cell3 volume = 2.842066937809649e-05 native^3.
```

5. Proposed 30 ft horizontal-axis rotor experiment

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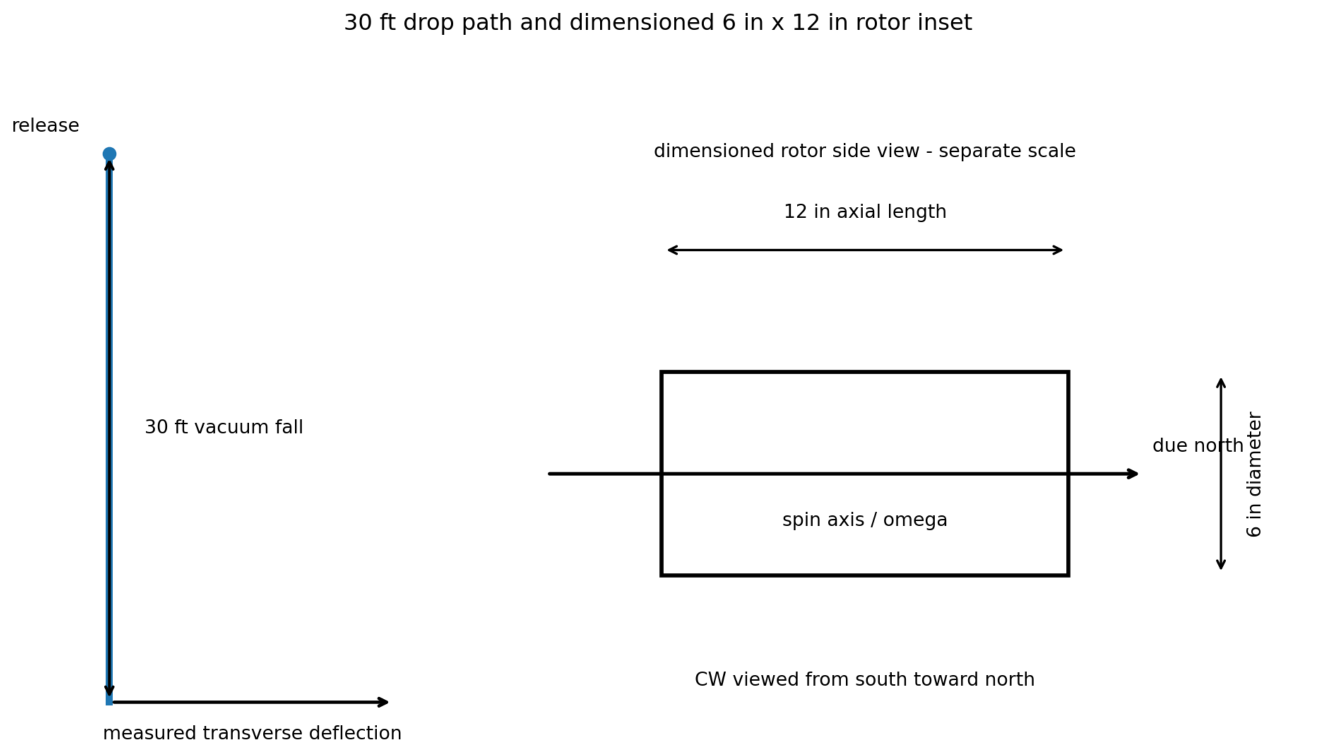

Figure 3. Dimensioned experimental geometry from the supplied values. The mass is intentionally not shown because no measured mass or steel alloy density was supplied.

The test body is a 6 in diameter, 12 in long steel cylinder. Its geometric volume is computed exactly as 0.00555999983 m^3. The axis is horizontal and points due north. Clockwise spin viewed from south toward north corresponds to a positive angular-velocity vector toward north. The vacuum drop distance is 30 ft (9.144 m).

The steel alloy and measured mass are not supplied. Accordingly, this report does not assign a density and does not publish a numerical K-Field force. The listed acceleration and deflection are the standalone-rotor maximum, for which the active rotating mass is the complete falling test body (m_r/M = 1). Any attached carriage, bearing, reflector, drive component, or release fixture that remains dynamically part of the falling body reduces the result by the measured active-mass fraction m_r/M.

Lock / RPM	omega (rad/s)	a_max (m/s^2)	a_max (micro-g)	30 ft time (s)	d_max (mm)	d_max (in)	Force
Source lock, 3600	376.991118	0.00539361965	549.99614	1.36559768	5.02916471	0.197998611	Not computable: mass absent
Source lock, 7200	753.982237	0.0107872393	1099.9923	1.36559768	10.0583294	0.395997221	Not computable: mass absent
Observer lock, 3600	376.991118	0.00478528943	487.96372	1.36559768	4.46194027	0.17566694	Not computable: mass absent
Observer lock, 7200	753.982237	0.00957057887	975.92744	1.36559768	8.92388055	0.35133388	Not computable: mass absent

At a fixed field-response lock, drop height, alignment, and active-mass fraction, d is exactly linear in RPM. Doubling 3600 rpm to 7200 rpm doubles acceleration, final transverse velocity, and deflection. Shortening or thinning a bare rotor leaves the standalone acceleration unchanged, but thinning a rotor on a fixed-mass carrier reduces m_r/M and therefore reduces the carrier-plus-rotor deflection.

6. Actual observer-baseline daily modulation for the due-north axis

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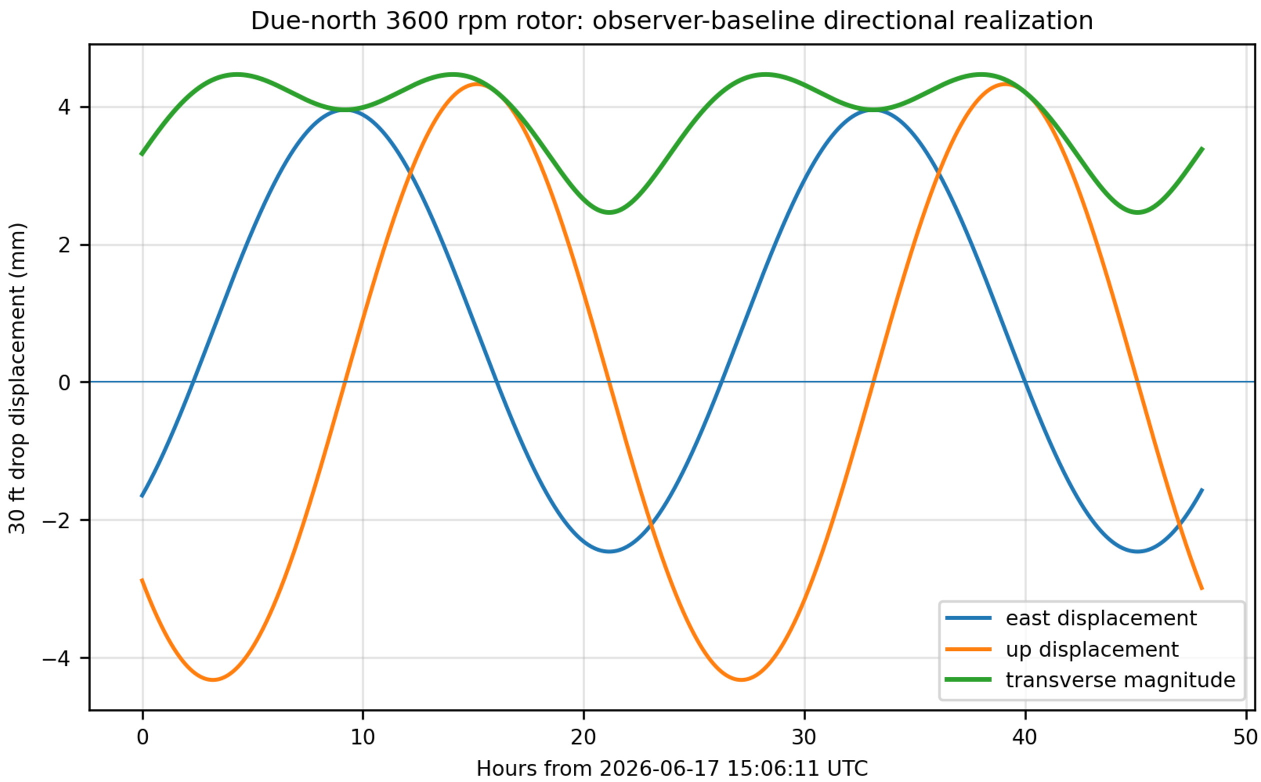

Figure 4. One-minute computation over 48 h for the specified due-north 3600 rpm axis and 30 ft fall. East and up are signed; magnitude is unsigned.

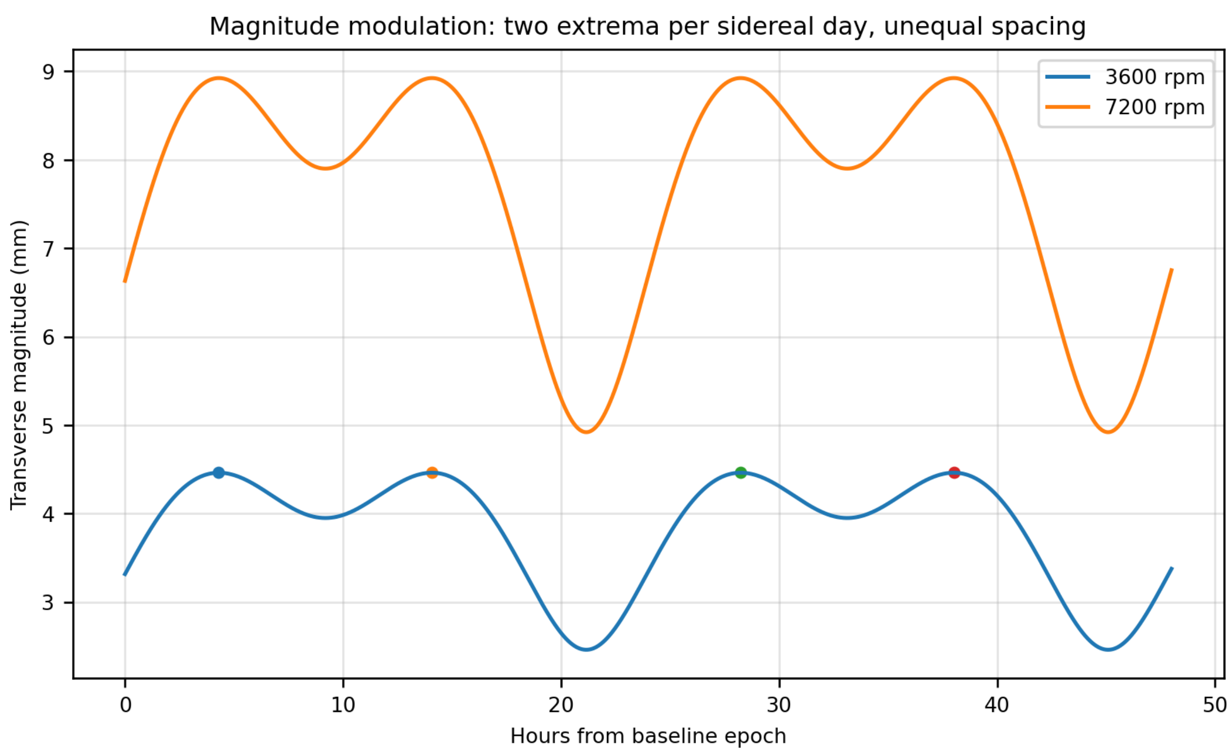

Figure 5. The 7200 rpm curve is exactly twice the 3600 rpm curve. The magnitude has two extrema per sidereal day, but their spacing is not exactly 12 h because the field vector has a nonzero offset relative to the rotating local north cone.

The signed east and up components repeat with the sidereal rotation of the laboratory. The unsigned magnitude reaches two maxima per sidereal day. In the first 48 h, the computed peak spacings alternate between approximately 9.78 h and

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14.15 h rather than forming a uniform 12 h sequence. The observer-locked 3600 rpm magnitude spans 2.460 to 4.462 mm; 7200 rpm doubles that span to 4.920 to 8.924 mm.

The first four one-minute-sampled maxima occur at 2026-06-17 19:24:11 UTC, 2026-06-18 05:11:11 UTC, 2026-06-18 19:20:11 UTC, and 2026-06-19 05:07:11 UTC. Spin reversal changes the signs of east and up displacement but leaves the magnitude unchanged. This distinction is essential: a signed channel is principally sidereal-day periodic, whereas an unsigned radial statistic can present two extrema per sidereal day.

Type	Epoch UTC	Hours from baseline	Magnitude (mm)
maximum	2026-06-17T19:24:11Z	4.300	4.461939
maximum	2026-06-18T05:11:11Z	14.083	4.461938
maximum	2026-06-18T19:20:11Z	28.233	4.461939
maximum	2026-06-19T05:07:11Z	38.017	4.461939
minimum	2026-06-18T00:17:11Z	9.183	3.949908
minimum	2026-06-18T12:15:11Z	21.150	2.460150
minimum	2026-06-19T00:14:11Z	33.133	3.949909
minimum	2026-06-19T12:12:11Z	45.100	2.460149

7. Literature survey and computation eligibility

The review follows an input-first method. Each paper is read for rotor mass, geometry, RPM, fall path or time, axis orientation, measurement channel, laboratory location, and timing information. A K-Field row is emitted only when the requested quantity is supported by those inputs. Maximum source-lock envelopes are used for historical comparisons because the papers generally omit exact drop epoch and observer state; these envelopes are not asserted to be the realized experimental values.

Study	Test type	Explicit mechanical data	K-Field computation status
J. Luo et al.	Double free-fall interferometer; rotating steel gyro versus nonrotating mass; nominal vertical rotor axis.	Rotating steel rotor mass 420.0 +/- 2.5 g; CCR mass 76.4 +/- 0.4 g; aluminum frame mass 159.4 +/- 0.9 g; total falling test-mass assembly 655.8 g nominal; rotor diameter about 55 mm; height about 32 mm; 17000 +/- 200 rpm; two 12 m tubes; approximately 10 m monitored path; 20-50 mPa.	Source-lock maximum rotor force and complete falling-assembly acceleration/displacement envelopes computed. Exact drop epoch and realized K-Field angle were not reported.
A. L. Dmitriev et al.	Closed falling container with internal horizontal-axis rotor system; speed up to 20000 rpm.	Paper states maximum RPM and approximately 14-15 min rotation time; rotor mass, dimensions, exact drop path, exact epoch, and full axis attitude are not supplied in the paper.	No K-Field force, acceleration, or displacement computed: rotor geometry/mass and drop trajectory are incomplete.
A. L. Dmitriev et al.	Vacuum aviation rotor in isolated container; horizontal-axis measurements over 20-400 Hz; 30 mm fall path.	Maximum 400 Hz (24000 rpm); runout 22 min; fall path 30 mm; 40 ms readout; 0.5-1 min sample interval; container 82 x 82 x 66 mm. Rotor mass and rotor geometry are absent.	No K-Field force, acceleration, or displacement computed: rotor geometry and mass are unavailable.
F. T. Han et al.	Proposed concentric hollow aluminum rotating masses in a 3.5 s double-capsule drop.	Inner: 27.1 g, radii 15.0/17.5 mm, length 39.92 mm, 2000 rpm. Outer: 69.7 g, radii 25.0/27.5 mm, length 64.37 mm, 20 rpm; speeds may be exchanged.	Source-lock force and individual-test-mass free-response acceleration envelopes computed from explicit mass, geometry, and RPM. No ballistic displacement is computed because the design keeps each mass electrostatically servo-controlled relative to the instrument frame. The K-Field force for a vertical spin axis is horizontal, so the proposed primary vertical channel is geometrically mismatched to that transverse force.
T. Liu et al.	Beijing drop-tower residual-acceleration characterization with single and double capsules.	Nearly 3.5 s free fall; double-capsule residual better than 2e-4 g; instrument accuracy better than 4e-5 g.	No K-Field computation: no rotating test mass, RPM, or rotor geometry.
J. Min et al.	Twenty three-axis drop-tower tests of a substitute inertial sensor.	Approximately 3.5 s fall; measured approximately 58 micro-g along fall direction and 13 micro-g horizontally in the most stable interval.	No K-Field computation: no rotating test mass, RPM, or rotor geometry.

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Study	Test type	Explicit mechanical data	K-Field computation status
T. J. Quinn and A. Picard et al.	Precision weighing of a freely spinning-down rotor.	330 g rotor, diameter 4.0 cm in Luo compilation, initial speed 8000 rpm.	No K-Field computation under the strict rule: the accessible source compilation gives mass, diameter, and RPM but not the complete rotor geometry.
I. Lorincz and M. Tajmar et al.	High-precision weight measurements with vertical and horizontal rotor orientations.	Brass rotor mass 112 g; speed above 25000 rpm in short runs and approximately 19000 rpm in long runs. Complete rotor dimensions are not supplied in the accessible paper text used here.	No K-Field computation under the strict rule: complete rotor geometry is unavailable.

8. Paper-by-paper technical review

8.1 J. Luo, Y. X. Nie, Y. Z. Zhang, and Z. B. Zhou, Phys. Rev. D 65, 042005 (2002).

Configuration. Double free-fall interferometer; rotating steel gyro versus nonrotating mass; nominal vertical rotor axis.

Source-reported data. Rotating steel rotor mass 420.0 +/- 2.5 g; CCR mass 76.4 +/- 0.4 g; aluminum frame mass 159.4 +/- 0.9 g; total falling test-mass assembly 655.8 g nominal; rotor diameter about 55 mm; height about 32 mm; 17000 +/- 200 rpm; two 12 m tubes; approximately 10 m monitored path; 20-50 mPa.

Source-reported result. No apparent differential acceleration at relative level 2e-6. Horizontal velocity difference estimated below 4.3 mm/s; approximately 6 mm lateral deviation over 10 m used as an upper estimate.

K-Field computation decision. Source-lock maximum rotor force and complete falling-assembly acceleration/displacement envelopes computed. Exact drop epoch and realized K-Field angle were not reported.

Quantity	Computed source-lock maximum
omega	1780.235837034 rad/s
Active rotating mass / total falling mass	0.4200 kg / 0.6558 kg
Active-mass fraction	0.640439158
Force on rotating rotor	0.010697346 N
Acceleration of complete falling assembly	0.016311902 m/s^2 (1663.351 micro-g)
Transverse displacement over reported 10 m monitored path	16.633511 mm

The paper measured vertical differential displacement and stated that its interferometer was nearly insensitive to horizontal motion. With a nominal vertical spin axis, the supplied K-Field cross product is transverse; therefore a vertical null is not a direct lateral null. The paper nevertheless supplies an indirect lateral constraint through beam overlap: approximately 6 mm deviation over a 10 m fall was used to bound horizontal velocity difference. Exact epoch and field angle are unavailable, so the 16.63 mm value above is a maximum envelope for the complete 655.8 g falling assembly, not a reconstructed prediction for the actual drops. No 12 m displacement is emitted because 12 m is the facility tube height, whereas the paper identifies approximately 10 m as the monitored fall path.

8.2 A. L. Dmitriev, E. M. Nikushchenko, and S. A. Bulgakova, arXiv:0907.2790 (2009).

Configuration. Closed falling container with internal horizontal-axis rotor system; speed up to 20000 rpm.

Source-reported data. Paper states maximum RPM and approximately 14-15 min rotation time; rotor mass, dimensions, exact drop path, exact epoch, and full axis attitude are not supplied in the paper.

Source-reported result. Reported average increase Delta g = 10 +/- 2 cm/s^2; stated that geographic north-south versus west-east orientation did not influence results and daily dependence was not observed.

K-Field computation decision. No K-Field force, acceleration, or displacement computed: rotor geometry/mass and drop trajectory are incomplete.

8.3 A. L. Dmitriev, arXiv:1101.4678 (2011).

Configuration. Vacuum aviation rotor in isolated container; horizontal-axis measurements over 20-400 Hz; 30 mm fall path.

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Source-reported data. Maximum 400 Hz (24000 rpm); runout 22 min; fall path 30 mm; 40 ms readout; 0.5-1 min sample interval; container 82 x 82 x 66 mm. Rotor mass and rotor geometry are absent.

Source-reported result. Reported stochastic frequency dependence, narrow extrema near 320 and 360 Hz, and predominantly negative changes of measured fall acceleration. The paper recommended high-time-resolution ballistic gravimetry using rotating or oscillating test bodies, but reported rotor data rather than a controlled oscillating-mass trajectory experiment.

K-Field computation decision. No K-Field force, acceleration, or displacement computed: rotor geometry and mass are unavailable.

8.4 F. T. Han, T. Y. Liu, X. X. He, and Q. P. Wu, Chin. Phys. Lett. 34, 100701 (2017).

Configuration. Proposed concentric hollow aluminum rotating masses in a 3.5 s double-capsule drop.

Source-reported data. Inner: 27.1 g, radii 15.0/17.5 mm, length 39.92 mm, 2000 rpm. Outer: 69.7 g, radii 25.0/27.5 mm, length 64.37 mm, 20 rpm; speeds may be exchanged.

Source-reported result. Conceptual design and uncertainty budget; no completed anomalous drop result. Sensitive x-axis is vertical during drop.

K-Field computation decision. Source-lock force and individual-test-mass free-response acceleration envelopes computed from explicit mass, geometry, and RPM. No ballistic displacement is computed because the design keeps each mass electrostatically servo-controlled relative to the instrument frame. The K-Field force for a vertical spin axis is horizontal, so the proposed primary vertical channel is geometrically mismatched to that transverse force.

Test mass	Mass	Geometry	RPM	Free-TM a_max	F_max	Ballistic displacement
Inner	27.1 g	r=15.0-17.5 mm; L=39.92 mm	2000	0.002996455 m/s^2	8.120394034e-05 N	Not computed: servo-controlled
Outer	69.7 g	r=25.0-27.5 mm; L=64.37 mm	20	0.000029965 m/s^2	2.088529388e-06 N	Not computed: servo-controlled

The force and free-test-mass acceleration columns are source-lock maximum envelopes for the individual rotating mass before electrostatic servo cancellation. They are not capsule center-of-mass accelerations. The published design keeps both masses electrostatically centered relative to the instrument frame during the drop, so a free ballistic displacement is not the measured observable and is not computed here. The intended observable is the differential control force or acceleration needed to maintain the common trajectory. The sensitive axis is vertical during the drop, while the supplied K-Field force for a vertical spin axis lies horizontally.

8.5 T. Liu, Q. Wu, B. Sun, and F. Han, Sci. Rep. 6, 31632 (2016).

Configuration. Beijing drop-tower residual-acceleration characterization with single and double capsules.

Source-reported data. Nearly 3.5 s free fall; double-capsule residual better than 2e-4 g; instrument accuracy better than 4e-5 g.

Source-reported result. Ordinary residuals associated with air drag, capsule rotation, accelerometer recovery, and tower airflow structure.

K-Field computation decision. No K-Field computation: no rotating test mass, RPM, or rotor geometry.

8.6 J. Min et al., npj Microgravity 7, 25 (2021).

Configuration. Twenty three-axis drop-tower tests of a substitute inertial sensor.

Source-reported data. Approximately 3.5 s fall; measured approximately 58 micro-g along fall direction and 13 micro-g horizontally in the most stable interval.

Source-reported result. 87 micro-g gain-switch transient, approximately 35 nm displacement vibration, approximately 26.56 Hz capsule resonance, and release disturbance.

K-Field computation decision. No K-Field computation: no rotating test mass, RPM, or rotor geometry.

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8.7 T. J. Quinn and A. Picard, Nature 343, 732-735 (1990).

Configuration. Precision weighing of a freely spinning-down rotor.

Source-reported data. 330 g rotor, diameter 4.0 cm in Luo compilation, initial speed 8000 rpm.

Source-reported result. Apparent mass changes were reduced by more than an order of magnitude and became insignificant after friction-couple and temperature corrections.

K-Field computation decision. No K-Field computation under the strict rule: the accessible source compilation gives mass, diameter, and RPM but not the complete rotor geometry.

The accessible compilation supplies rotor mass, diameter, and initial RPM, but not the axial dimension or another complete three-dimensional rotor specification. In accordance with the strict computation rule, no K-Field force, acceleration, or displacement is emitted for this study.

8.8 I. Lorincz and M. Tajmar, Measurement 73, 453-461 (2015).

Configuration. High-precision weight measurements with vertical and horizontal rotor orientations.

Source-reported data. Brass rotor mass 112 g; speed above 25000 rpm in short runs and approximately 19000 rpm in long runs. Complete rotor dimensions are not supplied in the accessible paper text used here.

Source-reported result. Precession and vibration artifacts identified; no anomaly within Delta m/m < 2.6e-6.

K-Field computation decision. No K-Field computation under the strict rule: complete rotor geometry is unavailable.

8.9 Oscillating test bodies: located evidence and missing experiment

Within the reviewed public source set, no completed drop-tower experiment was located that drove a mechanically oscillating test mass with a fully reported mass, geometry, amplitude, frequency, phase, drop trajectory, and signed lateral readout comparable to the proposed rotor test. Dmitriev 2011 explicitly recommended high-time-resolution ballistic gravimetry with rotating or oscillating test bodies [3], but the reported measurements used a rotor. Consequently, no K-Field acceleration or displacement is calculated for an oscillating-mass study. A valid future calculation would require the exact oscillator kinematics and the specific K-Field coupling law for linear oscillation; rotor RPM cannot be substituted for that missing law.

This absence is methodologically important. It prevents a numerical comparison between angular and linear actuation using public drop-tower data and means that the present literature-based quantitative matrix is limited to rotating bodies with explicit RPM and geometry.

9. Historical rotating-body comparison compiled by Luo et al.

Luo et al. tabulated earlier beam-balance and free-fall tests. The accessible compilation generally supplies mass, diameter, RPM, and in some rows moment-of-inertia or equivalent-radius quantities, but it omits the directly measured rotor axial dimension for the earlier studies. Luo et al. also states that some unknown table parameters were calculated under a uniform-composition assumption. Those assumed quantities are not used here. Under the strict rule adopted in this report, the earlier incomplete rows therefore receive no K-Field number. Luo et al. (2002) is the only row in this compilation with explicit rotating-rotor mass, complete falling-assembly mass, RPM, diameter, axial height, and a quantified monitored fall path, so it is the only historical row calculated here.

Study	Method	Source mass / diameter / RPM	Geometry and inertial-mass status	K-Field result
Hayasaka & Takeuchi (1989), rotor 1	beam balance	rotor 140.0 g; D=52.00 mm; 13000 rpm	diameter only; axial dimension unavailable	not computed under the strict rule: complete rotor geometry is unavailable
Hayasaka & Takeuchi (1989), rotor 2	beam balance	rotor 175.0 g; D=58.00 mm; RPM unavailable	diameter only; axial dimension unavailable	not computed: RPM is not explicitly available in the source row
Faller et al. (1990)	beam balance	rotor 451.0 g; D=51.00 mm; 6000 rpm	diameter only; axial dimension unavailable	not computed under the strict rule: complete rotor geometry is unavailable
Quinn & Picard (1990)	beam balance	rotor 330.0 g; D=40.00 mm; 8000 rpm	diameter only in the accessible compilation; axial dimension unavailable	not computed under the strict rule: complete rotor geometry is unavailable

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Study	Method	Source mass / diameter / RPM	Geometry and inertial-mass status	K-Field result
Nitschke & Wilmarth (1990)	beam balance	rotor 142.0 g; D=38.40 mm; 22000 rpm	diameter only; axial dimension unavailable	not computed under the strict rule: complete rotor geometry is unavailable
Imanishi et al. (1991)	beam balance	rotor 129.0 g; D=50.00 mm; 11000 rpm	diameter only; axial dimension unavailable	not computed under the strict rule: complete rotor geometry is unavailable
Hayasaka et al. (1997)	single free fall	rotor 175.0 g; D=58.00 mm; 18000 rpm	diameter only; axial dimension and exact fall path unavailable	not computed under the strict rule: complete rotor geometry is unavailable
Luo et al. (2002)	double free fall	rotor 420.0 g; D=54.90 mm; 17000 rpm; total falling mass=655.8 g	diameter and axial height explicitly reported; rotor, CCR, and frame masses reported	F_rotor,max=0.010697346 N; a_assembly,max=0.016311902 m/s^2; d_assembly,max(10 m)=16.633511 mm

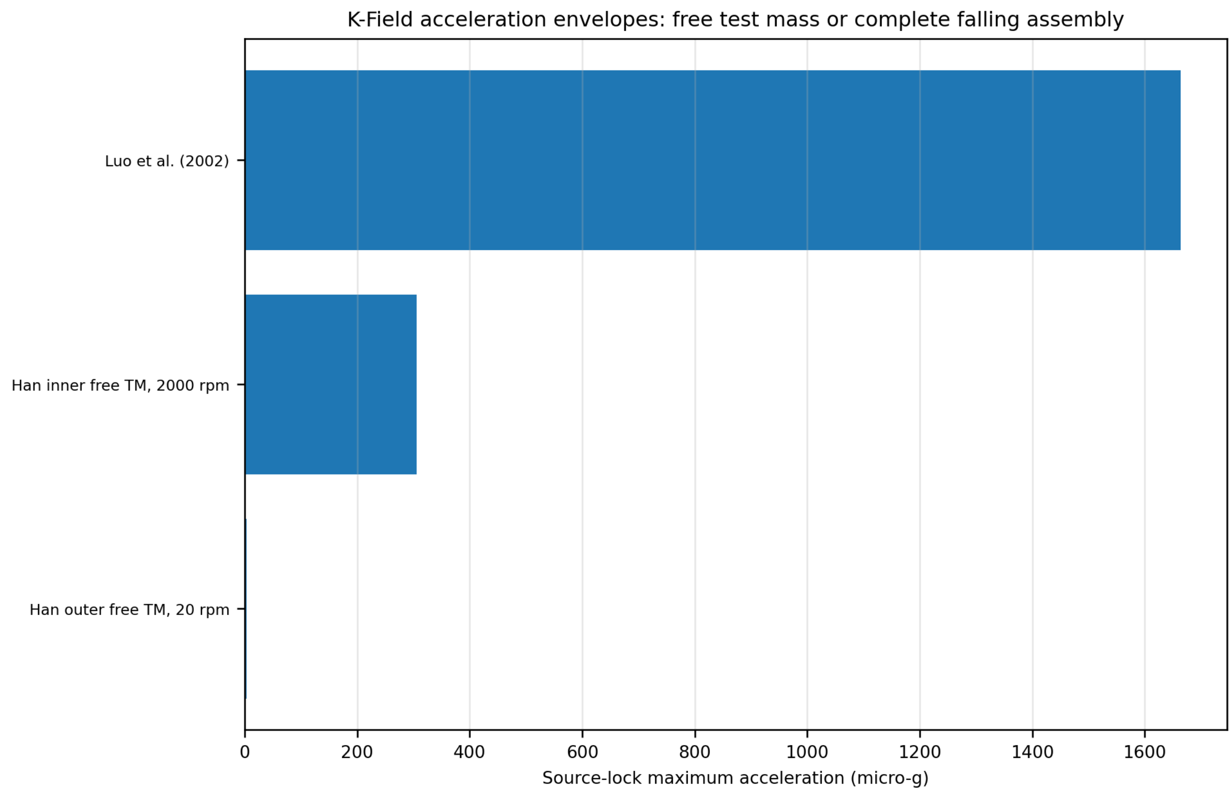

Figure 6. Source-lock acceleration envelopes only for configurations with explicit RPM, active rotating mass, and complete three-dimensional rotor geometry. Luo is the complete falling-assembly acceleration using the reported total assembly mass. Han values are individual-test-mass free-response normalizations before electrostatic servo cancellation. These are model comparisons, not reported residuals.

10. Reported residuals and ordinary error channels

Controlled studies do contain residual motion. The published interpretations generally connect those residuals to identifiable mechanical, environmental, optical, or control-system mechanisms. None of the reviewed papers establishes a confirmed unidentified 12-hour lateral-velocity or deflection residual.

Residual channel	Published manifestation	Experimental consequence
Rotor-frequency vibration	Luo: CCR vibration at rotor frequency; Lorincz/Tajmar: resonances and vibration artifacts	Can move the optical or weighing reference without changing center-of-mass trajectory
Frame or capsule rotation	Luo: approximately 1 Hz slow frame rotation; Liu/Han: normal and tangential apparent acceleration	Produces centrifugal, Euler, and Coriolis terms if sensor is offset
Release impulse and imbalance	Taiji: approximately 0.1 micrometre release displacement in first 0.2 s; capsule barycenter correction required	Creates initial lateral velocity that integrates directly into deflection
Residual gas and outgassing	Luo: asymmetric pump outgassing and residual gas; Liu/Min: air-drag effects	Generates direction-dependent acceleration and end-of-drop transients
Electrostatic control switching	Taiji: approximately 87 micro-g transient and approximately 35 nm displacement vibration	Produces a reproducible control artifact

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Residual channel	Published manifestation	Experimental consequence
Thermal and friction coupling	Quinn/Picard corrections; Lorincz/Tajmar apparatus characterization	Can create spin-speed-correlated apparent weight
Magnetic interaction	Luo estimated geomagnetic coupling; Han includes magnetic suspension before release	Requires spin-off magnetic controls and field mapping
Coriolis from release velocity	Luo estimated <54 microGal from horizontal velocity difference	Changes lateral trajectory without depending on rotor spin sign

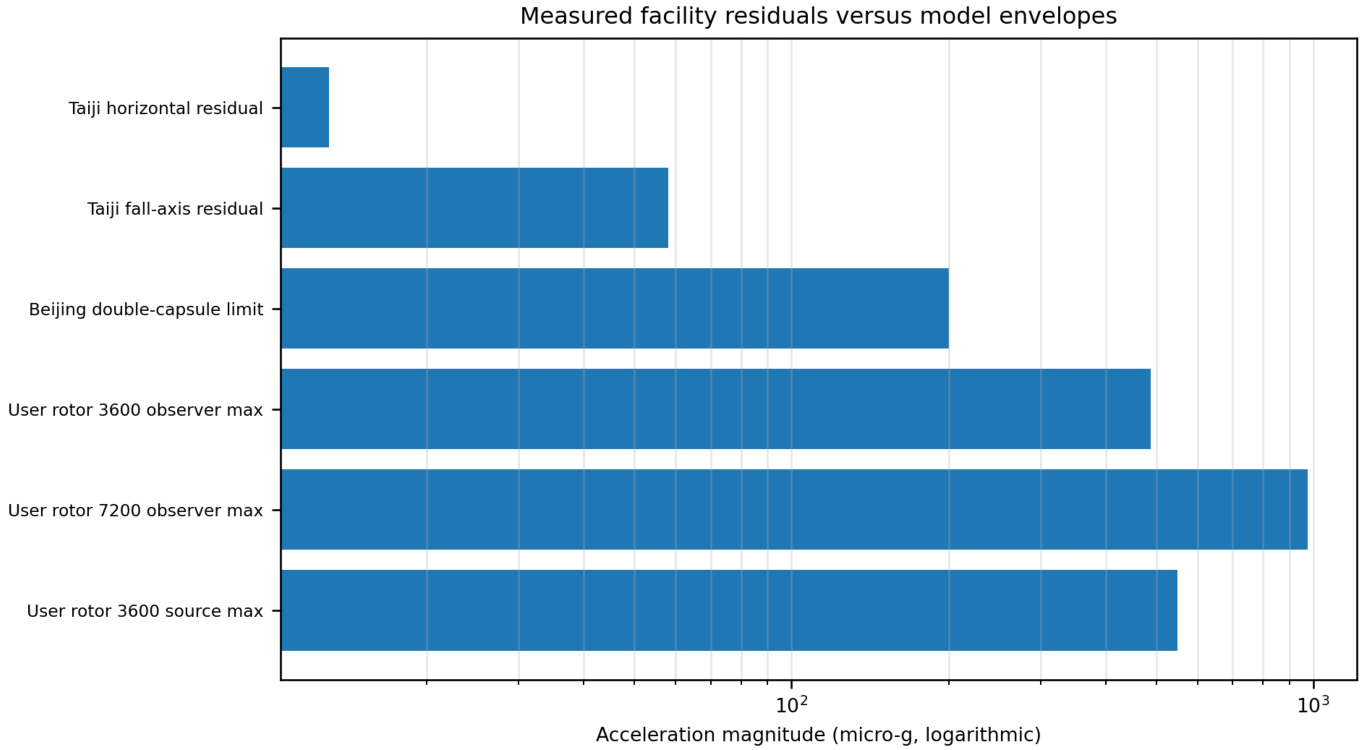

Figure 7. Published facility residuals are measured quantities; user-rotor bars are model envelopes. The categories are shown together only as an order-of-magnitude instrumentation comparison.

At the controlling observer lock, the 3600 rpm maximum acceleration is 487.96 micro-g and the 7200 rpm maximum is 975.93 micro-g. The Taiji study reported approximately 13 micro-g horizontally and 58 micro-g vertically in its most stable interval; the Beijing double-capsule study reported a bound better than 200 micro-g. These comparisons indicate that the proposed maximum signal is above published residual benchmarks, but they do not substitute for a rotor-specific noise budget.

11. The approximately 12-hour modulation question

The public record reviewed here does not contain a confirmed unexplained approximately 12-hour lateral-velocity or deflection modulation. Dmitriev et al. explicitly stated that daily dependence was not observed in their horizontal-axis experiment [2], but that paper did not provide a timestamped multi-day lateral trajectory series and did not supply the rotor geometry needed for a K-Field reconstruction. Other studies either averaged many drops, measured a vertical channel, or did not preserve exact UTC/local-sidereal metadata in the published record.

The actual baseline computation in Section 6 resolves the expected timing more precisely than a generic 12-hour statement. A laboratory-fixed signed displacement component repeats once per sidereal day. Its unsigned magnitude has two extrema per sidereal day; because the local north axis traces a cone and the field-response vector has a nonzero projection offset, adjacent extrema need not be separated by exactly half a sidereal day. For the baseline geometry, computed adjacent maxima are separated by approximately 9.78 h and 14.15 h in alternation.

A semidiurnal frequency by itself is not a unique identifier. NOAA lists the principal lunar M2 speed as 28.984104 deg per solar hour, the principal solar S2 speed as 30.0 deg per hour, and the lunisolar K2 speed as 30.0821373 deg per hour [15]. These correspond to periods of 12.420601 h, 12.000000 h, and 11.967235 h. The half-sidereal period is 11.967235 h,

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numerically coincident with K2 to the precision of the published constituent speed. Frequency fitting alone therefore cannot separate a semisidereal K-Field magnitude from a K2-period environmental or gravity term.

The M2-S2 beat period computed from the NOAA speeds is 14.765291 days. NOAA states that a minimum of 30 days of data is used to observe the majority of lunar and solar cycles and that a year is required to directly observe all standard constituents. A practical campaign must therefore exceed a few days and retain independent tidal, tilt, and gravity monitors.

A measurement program must retain signed east and up channels. Taking absolute value, radial magnitude, power, or squared displacement can conceal spin reversal and transform a one-sidereal-day signed vector into a two-extrema-per-day scalar pattern. The most discriminating statistic is the spin-odd vector

```
d_odd(t) = [d(+omega,t) - d(-omega,t)] / 2.
```

Tides, tower tilt, local gravity, and most release or environmental effects are spin-even and should cancel to first order in this channel. Rotor-coupled vibration, magnetic forces, and release changes that reverse with motor operation remain and must be measured independently.

Term	Angular speed (deg/h)	Period (h)	Interpretation
K-Field signed vector	15.041067	23.934470	One sidereal-day vector repeat
K-Field magnitude harmonic	30.082134	11.967235	Two-extrema scalar component
K2	30.0821373	11.967234788	Lunisolar semidiurnal constituent
S2	30.000000	12.000000000	Solar semidiurnal constituent
M2	28.984104	12.420601306	Principal lunar semidiurnal constituent

12. Experimental synthesis and recommended measurement architecture

A decisive drop-tower implementation should measure the full center-of-mass trajectory rather than infer lateral motion from beam intensity. Two orthogonal lateral interferometers or calibrated high-speed optical tracking should provide position, velocity, and attitude. A separate vertical channel records fall time and conventional acceleration. Rotor RPM and axis attitude must be recorded continuously from pre-release through capture.

Element	Required recorded quantity	Reason
Rotor state	RPM, signed spin direction, axis quaternion, imbalance spectrum	Determines K-Field prediction and separates vibration/precession
Observer state	UTC, UT1 correction if available, lat/lon/alt, local sidereal angle	Maps lab axis into ICRS and distinguishes sidereal timing
Trajectory	East, north, up position and velocity versus time	Separates signed vector deflection from radial magnitude
Controls	+RPM, -RPM, zero RPM, dummy rotor, release-only runs	Forms spin-odd and spin-even estimators
Environment	Pressure, magnetic field, temperature, tower tilt, capsule rotation	Models ordinary residuals documented in prior work
Analysis	Predicted vector projection before unblinding; sidereal and semisidereal terms	Avoids post hoc axis selection and recognizes unequal extrema spacing

The reviewed literature supports the feasibility of sensitive drop measurements, but it also shows that a vertical-only channel can miss a transverse force and that mechanically rotating hardware introduces strong spin-correlated artifacts. The proposed test is therefore not merely a repetition of prior equivalence-principle experiments; its distinguishing requirement is direct, signed, two-axis lateral trajectory reconstruction with explicit celestial-frame timing.

13. Findings

Finding	Supported conclusion
Closest direct free-fall null	Luo et al. supplied rotating-rotor mass, complete falling-assembly mass, rotor dimensions and RPM, and a sensitive vertical interferometer, but the instrument was nearly insensitive to horizontal motion.
Closest horizontal-axis nonzero reports	Dmitriev 2009/2011 reported vertical acceleration changes, but rotor geometry/mass and timestamped lateral trajectories are unavailable; no K-Field number is emitted.
Closest purpose-built rotating-mass drop design	Han et al. supplied complete concentric rotor dimensions, masses, RPM, and 3.5 s duration. It was a design study and its sensitive axis was vertical.
Best facility residual references	Liu 2016 and Min 2021 quantify ordinary drop-tower residuals and control transients without rotating test masses.

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Finding	Supported conclusion
Approximately 12 h residual	No confirmed unidentified 12-hour lateral modulation was found in the reviewed public research.
Model timing prediction	Signed components are sidereal-day periodic; magnitude has two extrema per sidereal day with unequal adjacent spacing for the baseline geometry.
User rotor maximum	Observer lock: 4.46194 mm at 3600 rpm and 8.92388 mm at 7200 rpm over 30 ft. Force remains uncomputed because mass was not supplied.
Thickness result	For a standalone rotor, the reduced law predicts unchanged acceleration at fixed RPM while force and angular momentum decrease. With any fixed-mass carrier, acceleration and deflection decrease as m_r/M; the arbitrarily thin limit therefore tends to zero for the complete assembly.

Appendix A. Complete Cell3 node register

Index	Class	lambda sum	ICRS x	ICRS y	ICRS z
(-1,-1,-1)	corner	-3	-0.010974902499961	0.007385983127423	0.039661911011939
(-1,-1,0)	edge center	-2	-0.011290092242635	-0.004742877881141	0.018199733726392
(-1,-1,1)	corner	-1	-0.011605281985309	-0.016871738889705	-0.003262443559155
(-1,0,-1)	edge center	-2	-0.009875529515800	0.013837068256769	0.022628660243911
(-1,0,0)	face center	-1	-0.010190719258473	0.001708207248205	0.001166482958364
(-1,0,1)	edge center	0	-0.010505909001147	-0.010420653760359	-0.020295694327182
(-1,1,-1)	corner	-1	-0.008776156531638	0.020288153386115	0.005595409475884
(-1,1,0)	edge center	0	-0.009091346274312	0.008159292377552	-0.015866767809663
(-1,1,1)	corner	1	-0.009406536016985	-0.003969568631012	-0.037328945095210
(0,-1,-1)	edge center	-2	-0.000784183241488	0.005677775879218	0.038495428053575
(0,-1,0)	face center	-1	-0.001099372984162	-0.006451085129346	0.017033250768028
(0,-1,1)	edge center	0	-0.001414562726835	-0.018579946137910	-0.004428926517519
(0,0,-1)	face center	-1	0.000315189742674	0.012128861008564	0.021462177285547
(0,0,0)	body center	0	0.000000000000000	0.000000000000000	0.000000000000000
(0,0,1)	face center	1	-0.000315189742674	-0.012128861008564	-0.021462177285547
(0,1,-1)	edge center	0	0.001414562726835	0.018579946137910	0.004428926517519
(0,1,0)	face center	1	0.001099372984162	0.006451085129346	-0.017033250768028
(0,1,1)	edge center	2	0.000784183241488	-0.005677775879218	-0.038495428053575
(1,-1,-1)	corner	-1	0.009406536016985	0.003969568631012	0.037328945095210
(1,-1,0)	edge center	0	0.009091346274312	-0.008159292377552	0.015866767809663
(1,-1,1)	corner	1	0.008776156531638	-0.020288153386115	-0.005595409475884
(1,0,-1)	edge center	0	0.010505909001147	0.010420653760359	0.020295694327182
(1,0,0)	face center	1	0.010190719258473	-0.001708207248205	-0.001166482958364
(1,0,1)	edge center	2	0.009875529515800	-0.013837068256769	-0.022628660243911
(1,1,-1)	corner	1	0.011605281985309	0.016871738889705	0.003262443559155
(1,1,0)	edge center	2	0.011290092242635	0.004742877881141	-0.018199733726392
(1,1,1)	corner	3	0.010974902499961	-0.007385983127423	-0.039661911011939

Appendix B. Reproducible numerical identities

Identity	Computed value
Node reconstruction max error	0.000e+00
Source q_g versus JSON q_g absolute difference	0.000e+00
Observer g reconstruction	G_phase multiplied by observer ICRS velocity
30 ft fall time	1.365597675120 s
Observer max 3600 rpm displacement	0.004461940274 m

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Identity	Computed value
Observer max 7200 rpm displacement	0.008923880549 m
Source max 3600 rpm displacement	0.005029164711 m

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